

Ornamentation Lattices and Intreeval Hypergraphic Lattices

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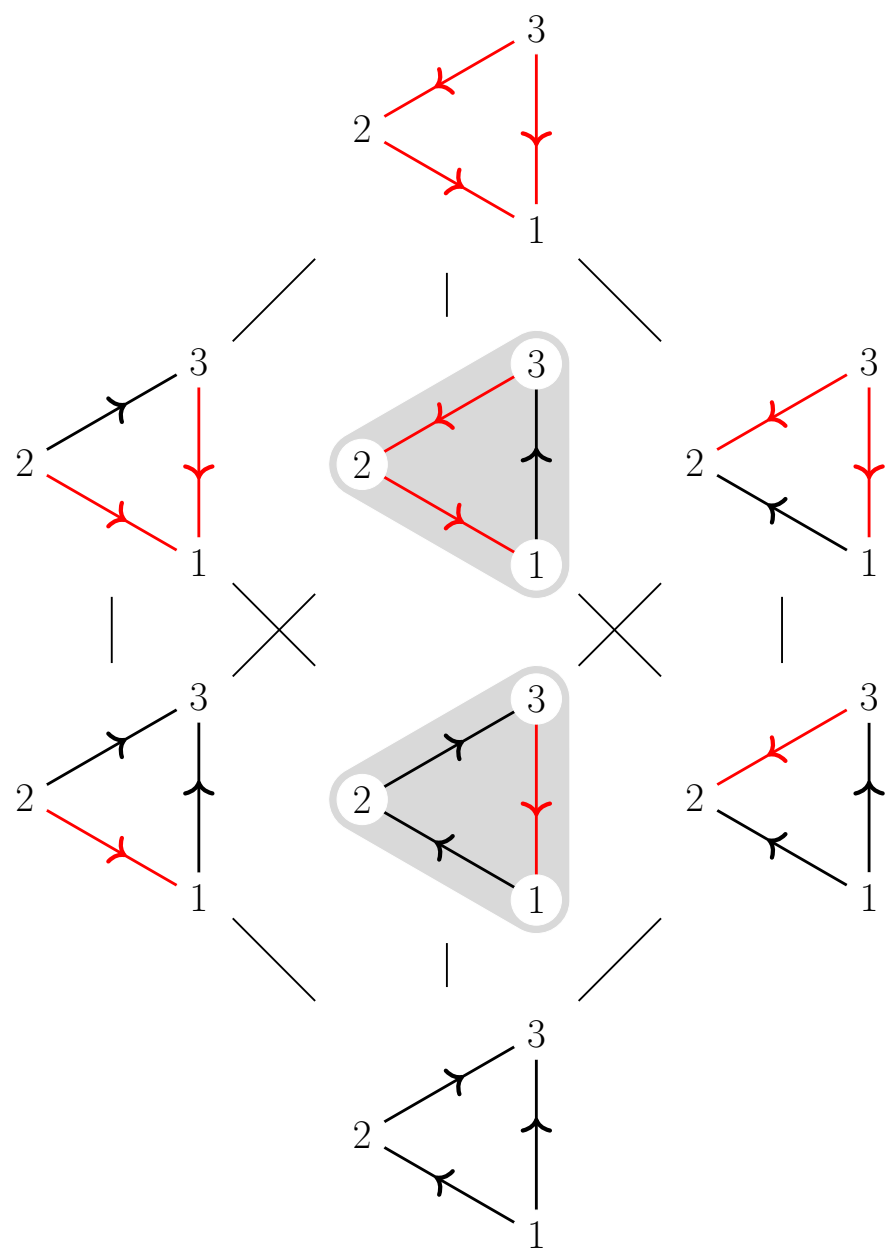
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Reorientations

Reorientation lattice $\mathcal{R}(D)$: Boolean lattice of reorientations, ordered by inclusion of subsets of reversed arrows.

Acyclic reorientation poset $\mathcal{AR}(D)$: Subposet of $\mathcal{R}(D)$.



Ornamentations

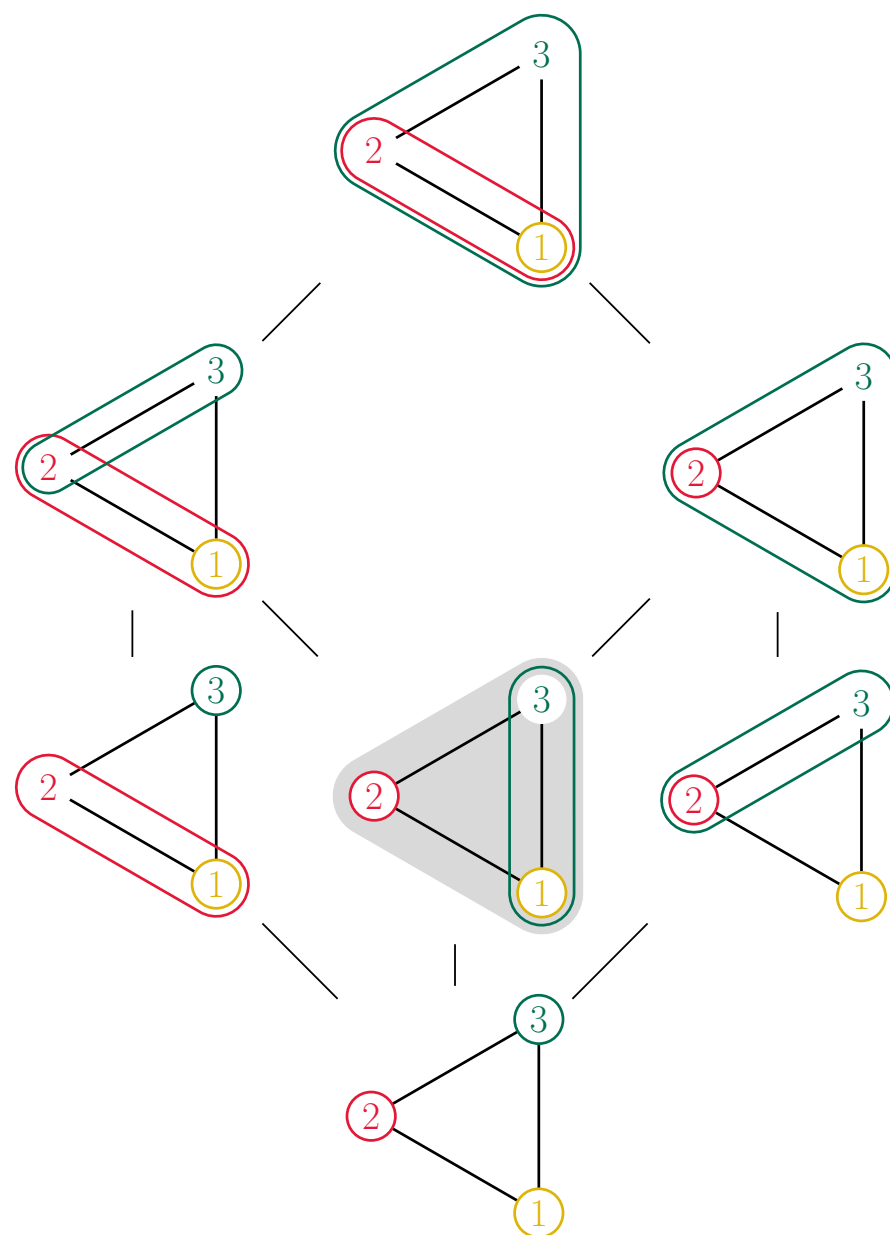
Ornament $O(v)$: Subset U such that there is a path from any $u \in U$ to v in the subgraph of D induced by U .

Ornamentation: Map O which assigns an ornament $O(v)$ at each vertex $v \in V$ such that $u \in O(v) \implies O(u) \subseteq O(v)$ for all $u, v \in V$.

Ornamentation lattice $\mathcal{O}(D)$: Lattice of ornamentations under componentwise inclusion, meaning $O_1 \leq O_2 \iff O_1(v) \subseteq O_2(v)$ for all $v \in V$.

Acyclic ornamentation: If there exists reorientation $R \in \mathcal{AR}(D)$ such that $\mathbf{AO}_R = O$.

Acyclic ornamentation poset $\mathcal{AO}(D)$: Subposet of $\mathcal{O}(D)$.



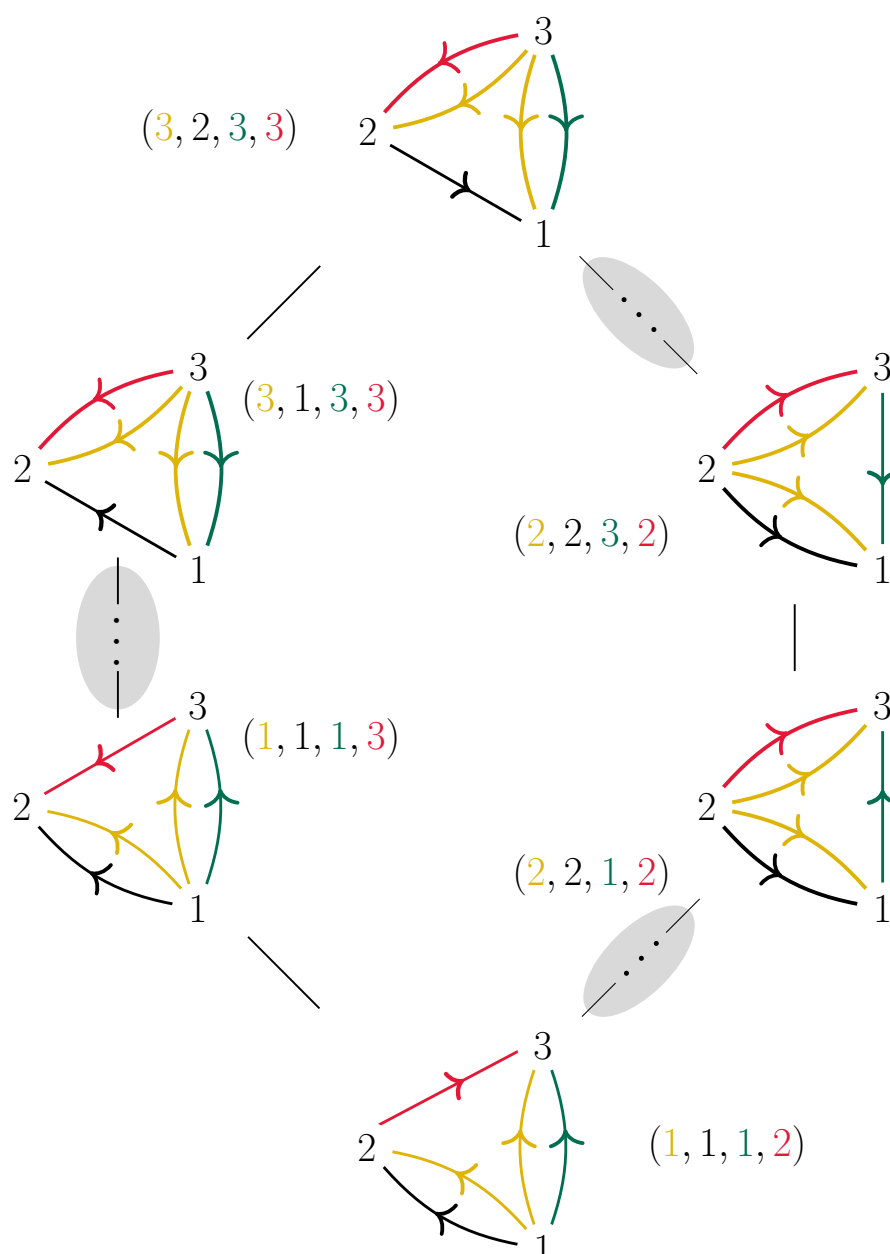
Sourcings

Sourcing S : $S: \mathbb{H} \rightarrow V$ such that $S(H) \in H$ for all $H \in \mathbb{H}$.

Sourcing lattice $\mathcal{S}(\mathbb{H})$: Lattice of sourcings ordered componentwise, meaning $S_1 \leq S_2 \iff S_1(H) \leq S_2(H)$ for all $H \in \mathbb{H}$.

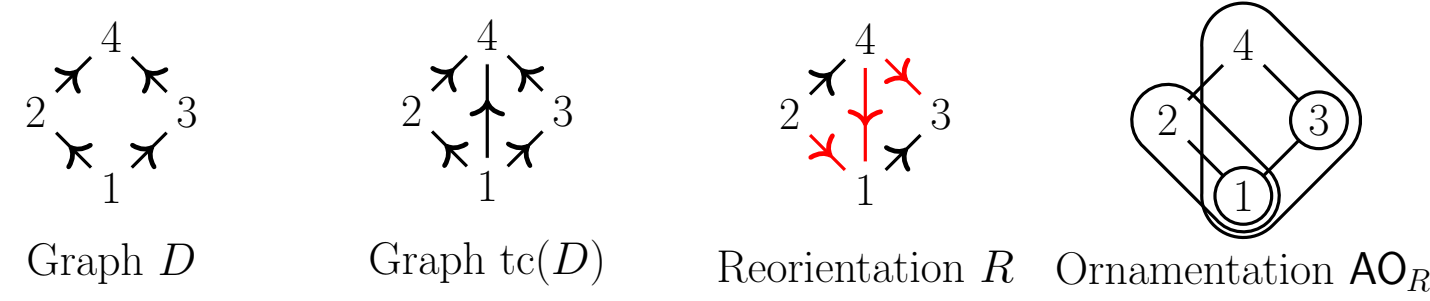
Acyclic sourcing: If no $H_0, \dots, H_k$ such that $S(H_{i-1}) \in H_i \setminus \{S(H_i)\}$ for all $i \in [k]$ and $S(H_k) \in H_0 \setminus \{S(H_0)\}$.

Acyclic sourcing poset $\mathcal{AS}(\mathbb{H})$: Subposet of $\mathcal{S}(\mathbb{H})$.

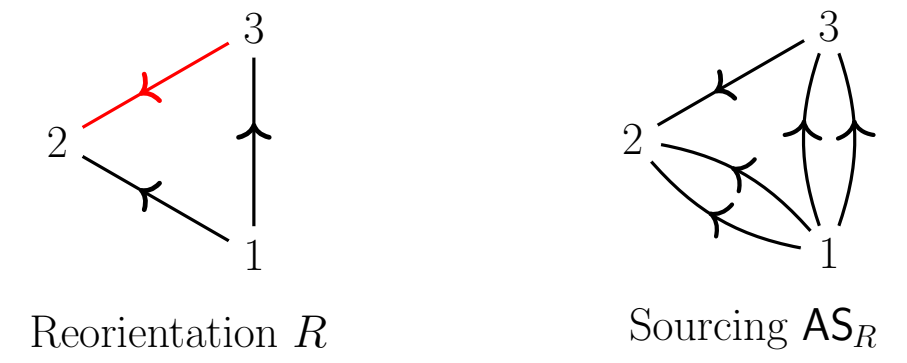


Maps

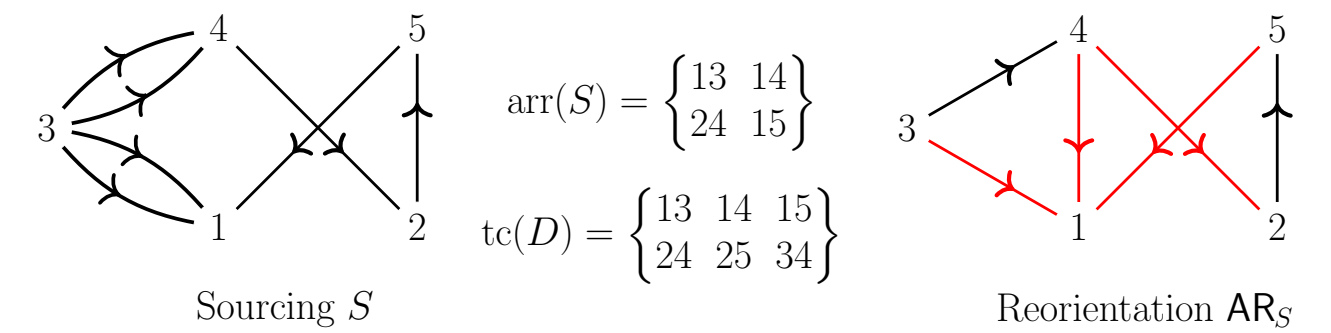
Map $R \mapsto \mathbf{O}_R$: Associates to each vertex $v \in V$ the inclusion maximal ornament at v contained in the subset of vertices $u \in V$ with a directed path to v using only the reoriented arrows of R .



Map $R \mapsto \mathbf{AS}_R$: Associates to each path $P \in \mathbb{P}$ the source $\mathbf{AS}_R(P)$, which is the vertex $u \in P$ such that $(u, v) \in R$ for all $v \in P \setminus \{u\}$.



Map $S \mapsto \mathbf{AR}_S$: The reoriented arrows are $\text{tc}(\text{arr}(S)) \cap \text{tc}(D)$.
 $\text{arr}(S) = \{(u, v) | \exists P \in \mathbb{P}(D) \text{ with } u \in P, v = S(P)\}$.



Map $S \mapsto \mathbf{O}_S$: Associates to each vertex $v \in V$ the inclusion maximal ornament at v contained in the subset of vertices $u \in V$ with a directed path to v using only the arrows of $\text{rev}(S)$.
 $\text{rev}(S) = \{(u, v) | \exists P = u, \dots, v \in \mathbb{P}(D) \text{ with } S(P) = v\}$.

Theorem [ABGPS25]:

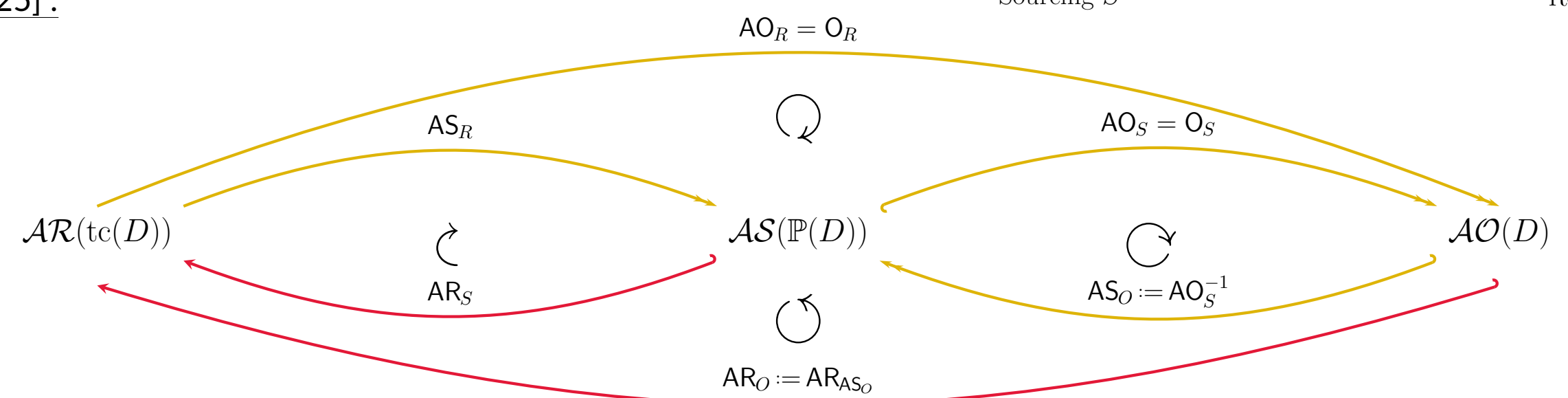


Figure 1. The maps in yellow are order preserving while those in red are not. The symbol $\curvearrowright$ means a commuting diagram, the symbol $\curvearrowleft$ means that one map is a section of the other.

MacNeille completions

MacNeille completion of a poset P : Smallest Lattice L such that there is an order-embedding $f: P \rightarrow L$.

Theorem [ABGPS25]: For a directed tree T ,

- the transitively biclosed reorientation lattice $\mathcal{R}^{\text{bi}}(\text{tc}(T))$ is the MacNeille completion of $\mathcal{AR}(\text{tc}(T))$,
- the ornamentation lattice $\mathcal{O}(T)$ is the MacNeille completion of $\mathcal{AO}(T) \simeq \mathcal{AS}(\mathbb{P}(T))$.

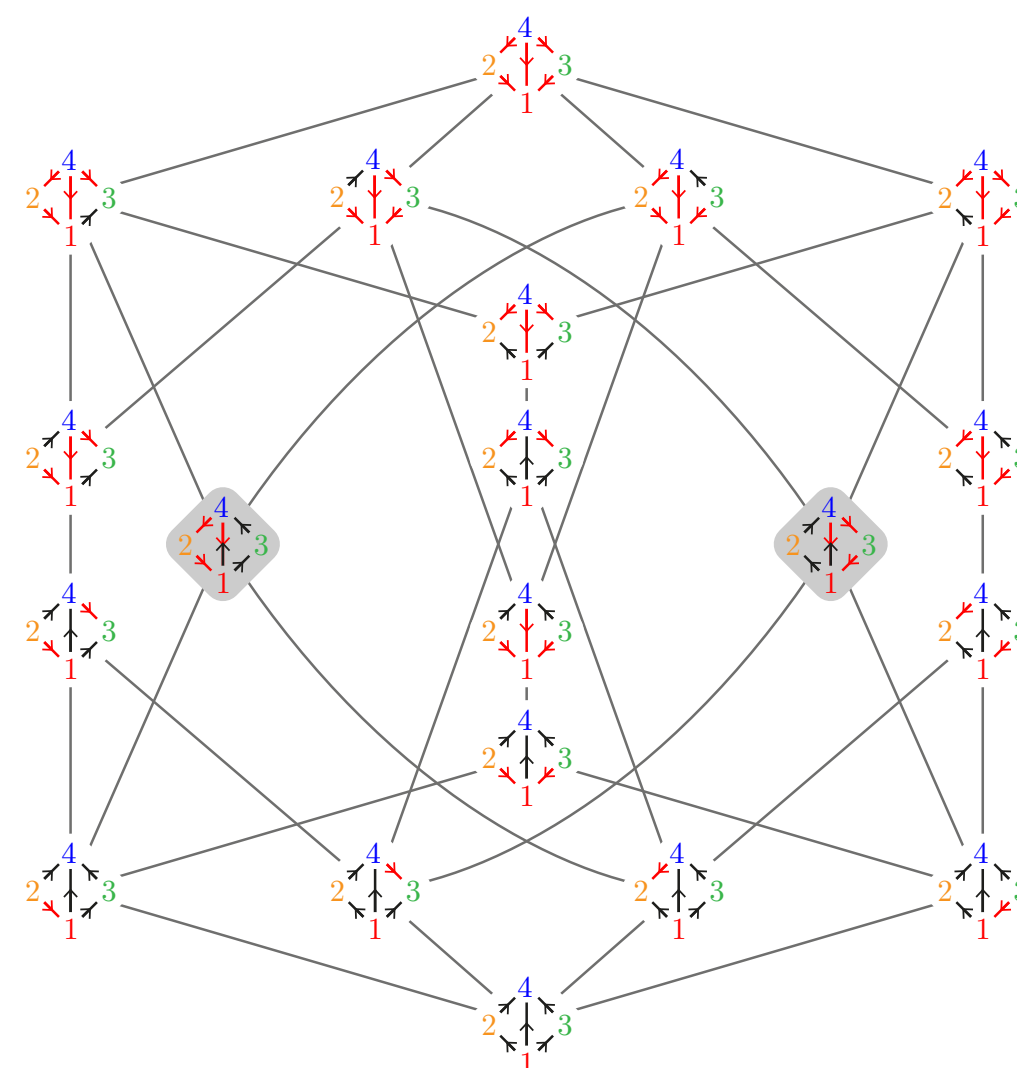


Figure 2. The MacNeille completion of $\mathcal{AR}(\text{tc}(\diamond))$.

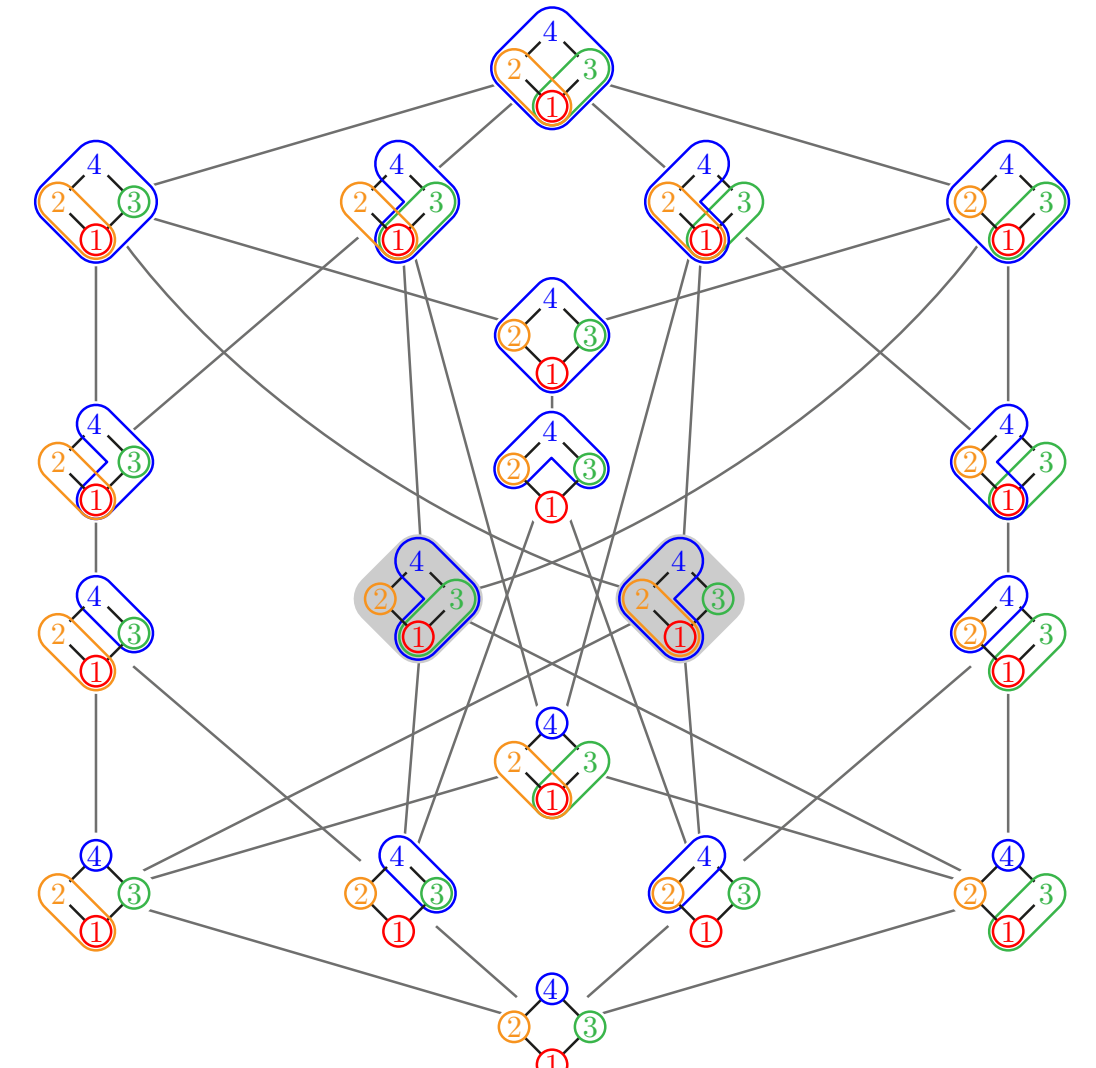


Figure 3. The MacNeille completion of $\mathcal{AO}(\diamond)$.

Unstarred and rooted trees T

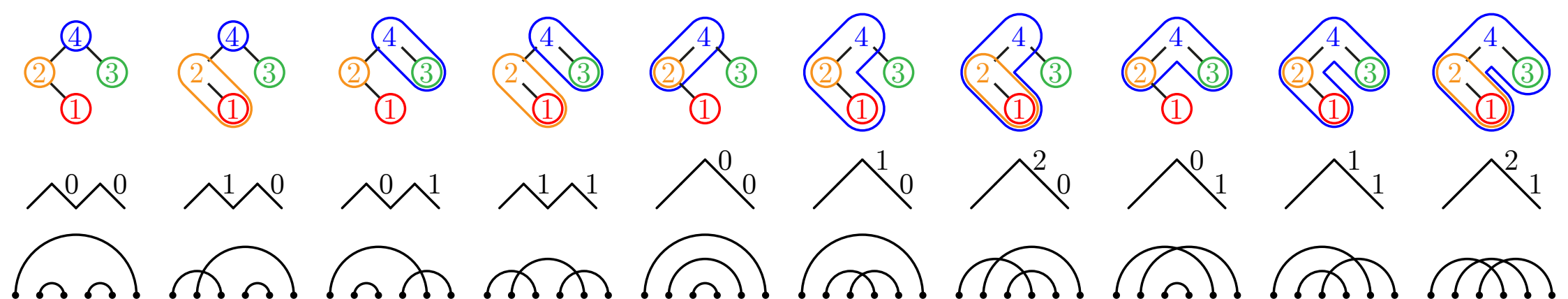
Starred three: If $\text{tc}(T)$ contains an induced alternating cycle, **unstarred** otherwise.

Theorem [ABGPS25]: For an unstarred increasing tree T ,

- $\mathcal{AR}(\text{tc}(T))$, $\mathcal{AS}(\mathbb{P}(T))$, and $\mathcal{AO}(T)$ are all lattices,
- all ornamentations of T are in fact acyclic, and so $\mathcal{AS}(\mathbb{P}(T)) \simeq \mathcal{AO}(T) = \mathcal{O}(T)$,
- $\mathcal{O}(T)$ is a lattice quotient of $\mathcal{AR}(\text{tc}(T))$,
- $\mathcal{O}(T)$ is isomorphic to the transitive closure of the graph of the hypergraphic polytope $\Delta_{\mathbb{P}(T)}$ oriented by $\mathbf{w}$.

Proposition [ABGPS25]: The n -comb is the rooted tree obtained by attaching n nodes to each node of a n -path. The following sets are in bijection:

- the (acyclic) ornamentations of the n -comb,
- the labeled Dyck paths of semilength n ,
- the indecomposable perfect matchings of $[2n + 2]$.

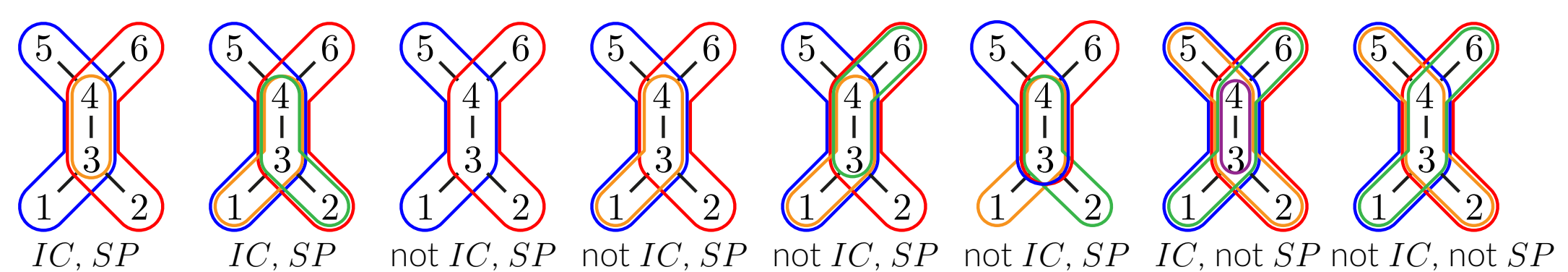


Intreeval hypergraphic lattices

Intreeval hypergraph: Subhypergraph of $\mathbb{P}(T)$.

Path intersection closed: If for all $I, J \in \mathbb{I}$ with $|I \cap J| > 1$ such that $\min(I) <_T \min(I \cap J)$ and $\max(J) >_T \max(I \cap J)$, there exists $K \in \mathbb{I}$ such that $I \cap J \subseteq K \subseteq [\min(J), \max(I)]_T$.

Star sparse: If, for all $u \leq_T v$, the (undirected) graph $X(T, u, v, \mathbb{I})$ on the incoming edges u_i at u and the outgoing edges v_j at v , with (u_i, v_j) an edge if and only if $u_i, v_j \in I$, is acyclic.



Theorem [ABGPS25]: The acyclic sourcing poset $\mathcal{AS}(\mathbb{I})$ is a lattice if and only if $\mathbb{I}$ is path intersection closed and star sparse.